

A photograph of a server room with blue ambient lighting. Several server racks are visible, with some doors open, revealing internal components. The room has a high ceiling with exposed ductwork and cables.

Virtualization

Brett Johnson

A photograph of a server room with blue ambient lighting. Rows of server racks are visible, with some racks having glass doors that show internal components. A semi-transparent white rectangular box is overlaid in the center of the image, containing the text "What is virtualization?".

What is virtualization?



What is virtualization?

A process that allows a computer to share its hardware resources with multiple digitally separated environments.

Running a computer in a Computer

What is virtualization?

- Provides an abstract layer between physical hardware (Host) and the virtual environment (VMs)
- Physical resources such as CPU, RAM, Storage, and Network usage are shared between multiple VMs
- Provides the ability to easily build and destroy different VMs without affecting the rest of the system

A photograph of a server room with blue ambient lighting. In the center, there is a semi-transparent white rectangular box containing the word "Hypervisors" in a bold, dark blue font. The background shows rows of server racks with some open doors and visible internal components.

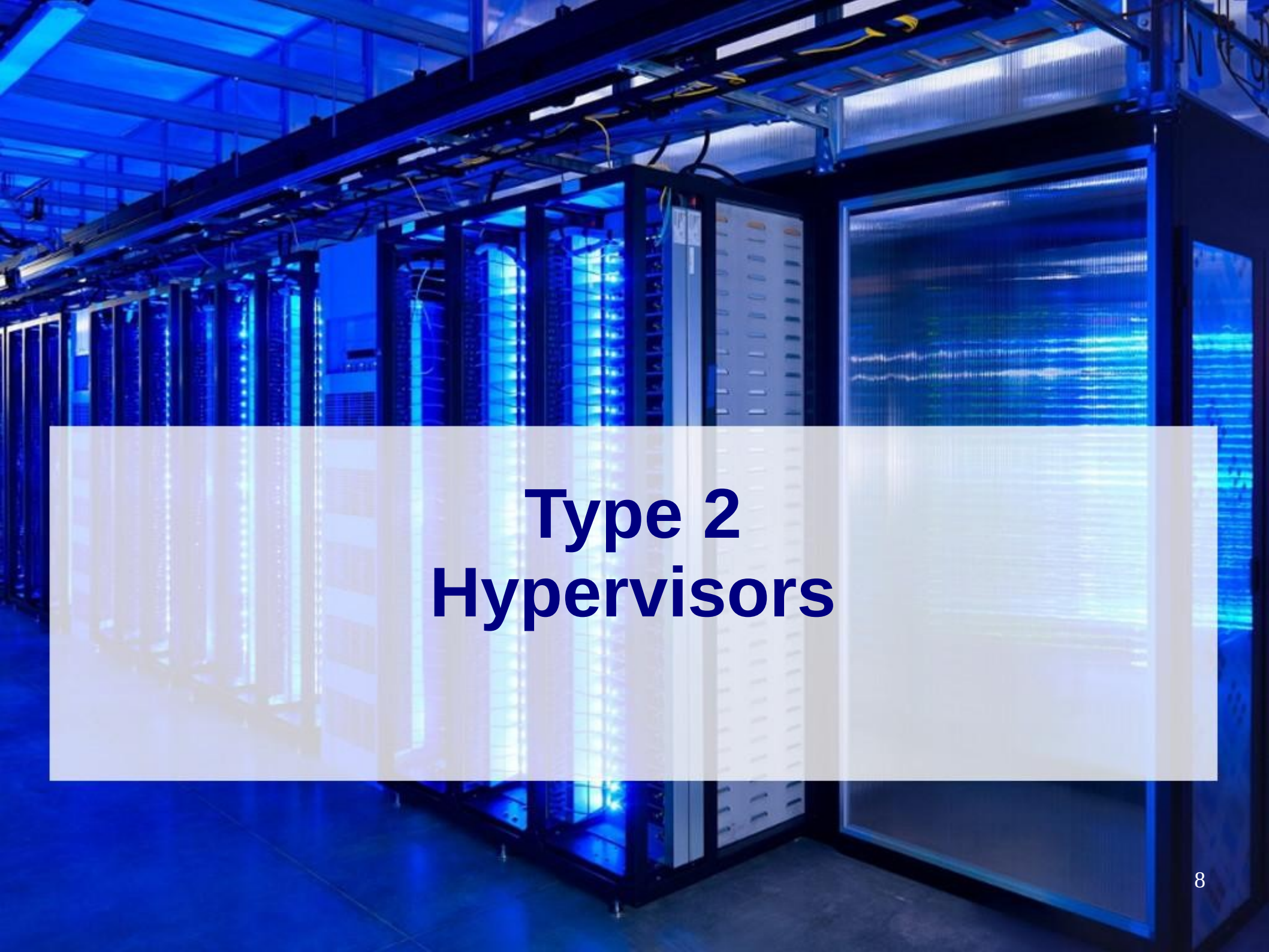
Hypervisors

Hypervisors

- Software that allows multiple VMs to run a single Host
- Manage the physical resources allocated to VMs
- Ensure that each VM is completely isolated

Hypervisors

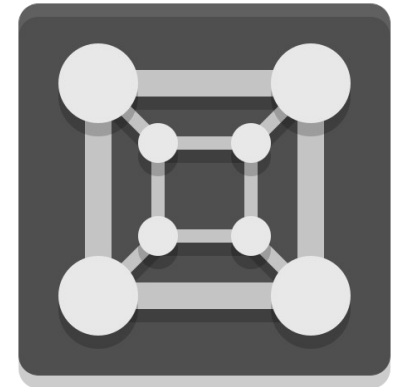
- Type 1 (Bare-metal or Native)
 - Runs directly on physical hardware
 - Is the Host Operating System (OS)
- Type 2 (Hosted)
 - Runs on top of a host OS

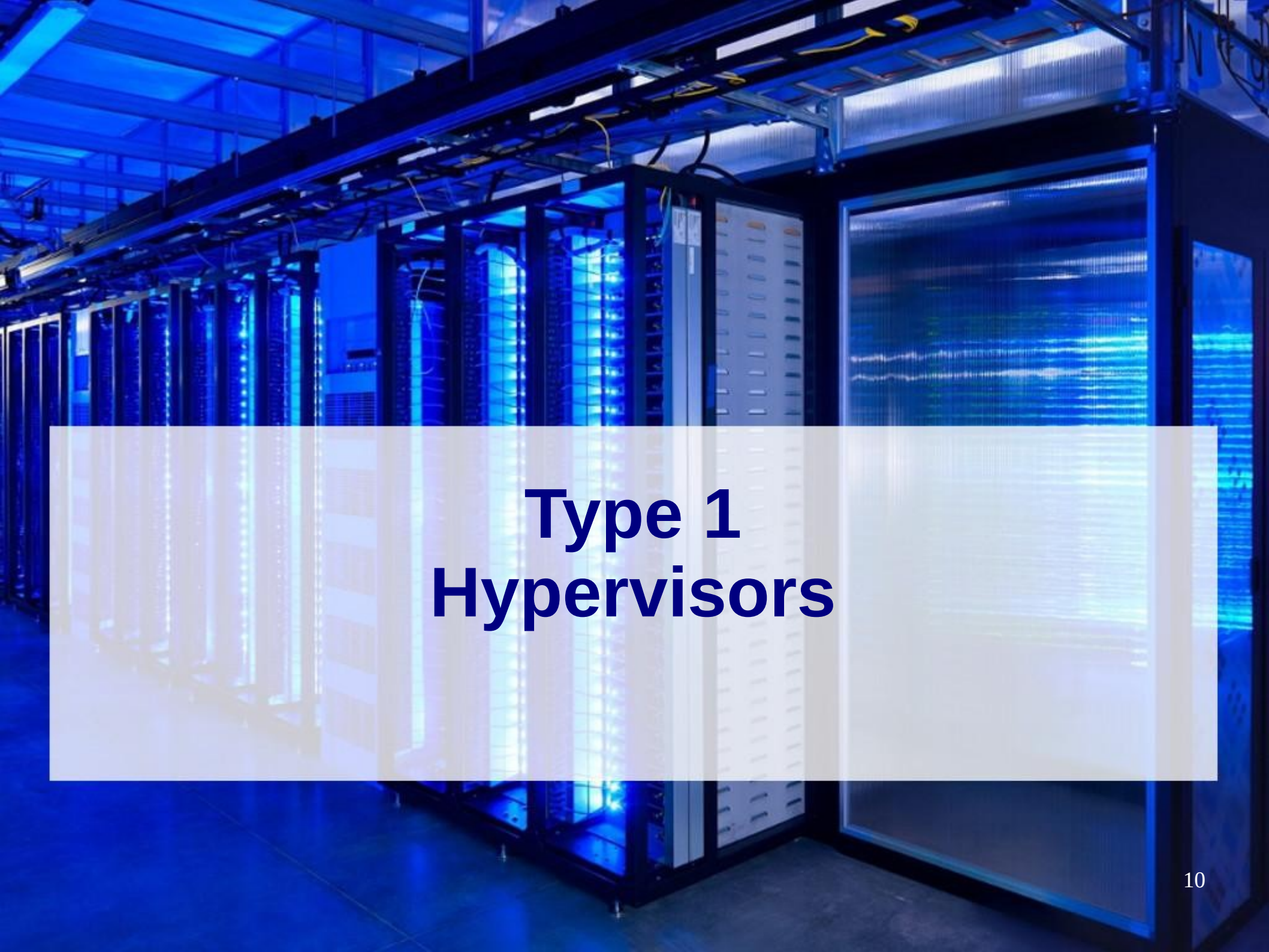
A photograph of a server room with blue ambient lighting. Rows of server racks are visible, with some racks having glass doors that show internal components. A semi-transparent white rectangle is overlaid in the center of the image, containing the text 'Type 2 Hypervisors'.

Type 2 Hypervisors

Type 2 Hypervisors

- Host is responsible for managing resources
- Great testing environments



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Type 1 Hypervisors

Type 1 Hypervisors

- Is the OS
- Server applications
 - Data centers
 - High Performance
 - Run other Services



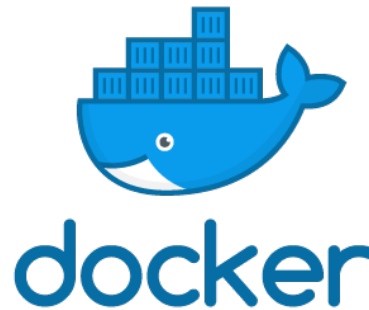
A photograph of a server room with blue ambient lighting. Rows of server racks are visible, with some racks having glass doors that show internal components. A semi-transparent white rectangular box is overlaid in the center of the image, containing the word "Containerization" in a bold, dark blue font.

Containerization

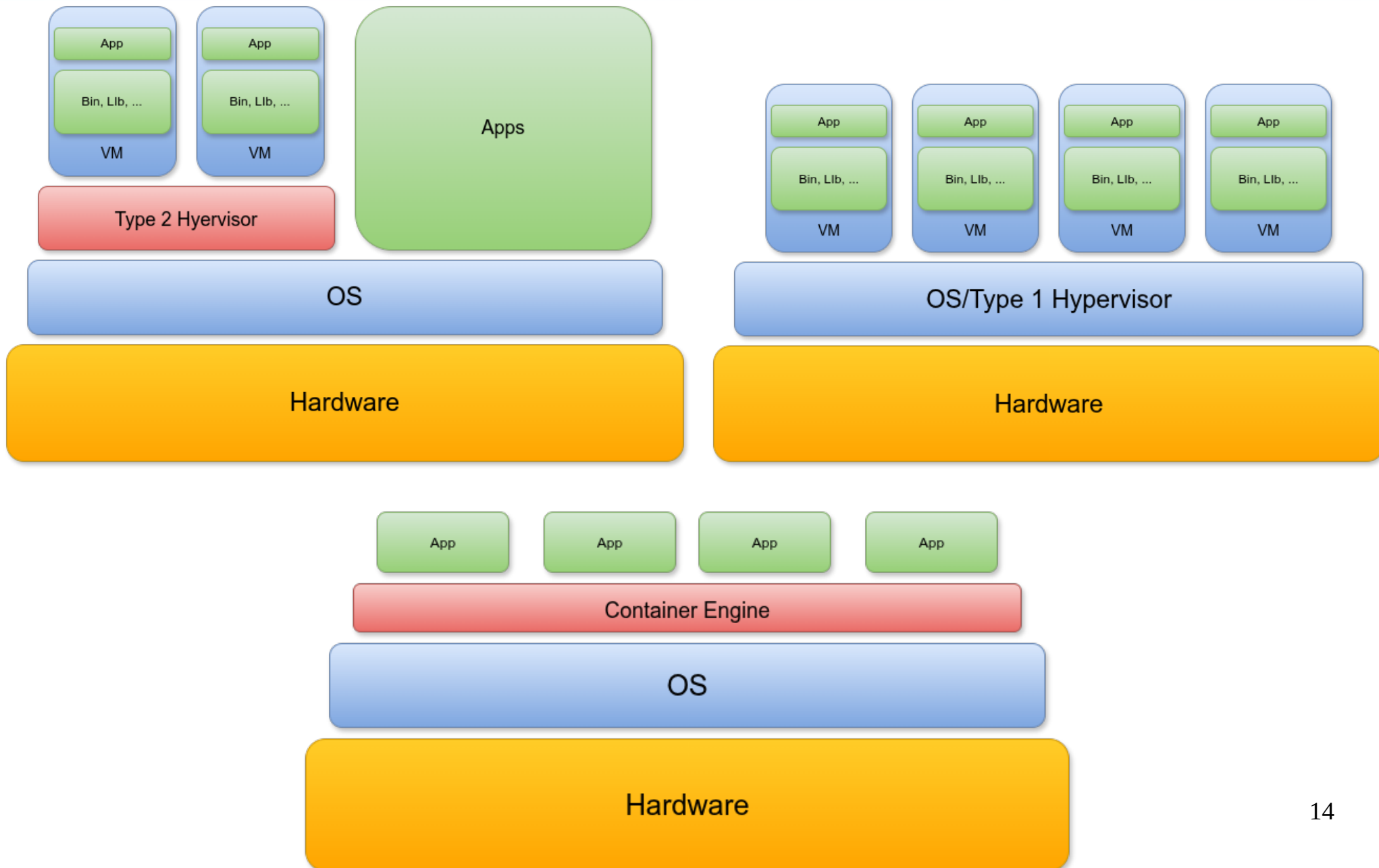
Containerization

- A way of packaging software and its dependencies into a single, portable unit
 - Containers

- Lightweight
- Still isolated
- Very Portable
- Easy to Scale



Hypervisors

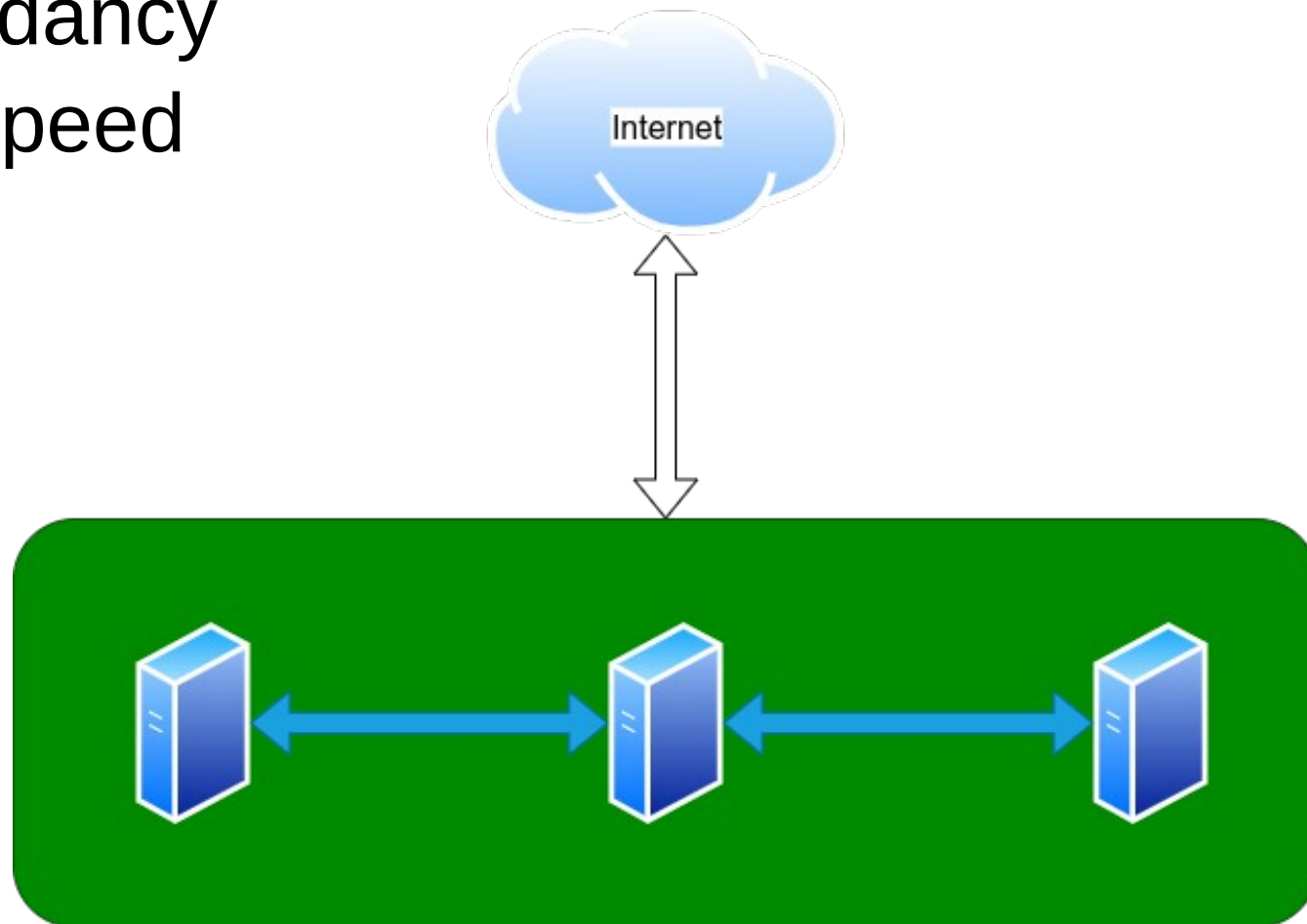


A photograph of a server room with blue ambient lighting. Rows of server racks are visible, with some racks having glass doors that show internal components. A semi-transparent white rectangle is overlaid in the center of the image, containing the text "Clusters & High Availability".

Clusters & High Availability

Clusters & HA

- Multiple hosts working together
- Redundancy
- High Speed



Clusters & HA

- CEPH
 - Should use 3 or more hosts
 - Each host gives its 'opinion'
 - Tie breaking vote wins

A photograph of a server room with blue ambient lighting. Rows of server racks are visible, with some racks having glass doors that show internal components. A semi-transparent white rectangular box is overlaid in the center of the image, containing the word "Applications" in a bold, dark blue font.

Applications

Applications

- Lab Environments (Build & Destroy)
- Sand-boxed Testing (Malware)
- Application Security (Flatpak)
- Services Ex: Nextcloud, PiHole, Website

A photograph of a server room with blue ambient lighting. Rows of server racks are visible, with some racks having glass doors that show internal components. A semi-transparent white rectangular box is overlaid in the center of the image, containing the text "Getting Started" in a bold, dark blue font.

Getting Started

VirtualBox

Installing VirtualBox

<https://www.virtualbox.org/wiki/Downloads>

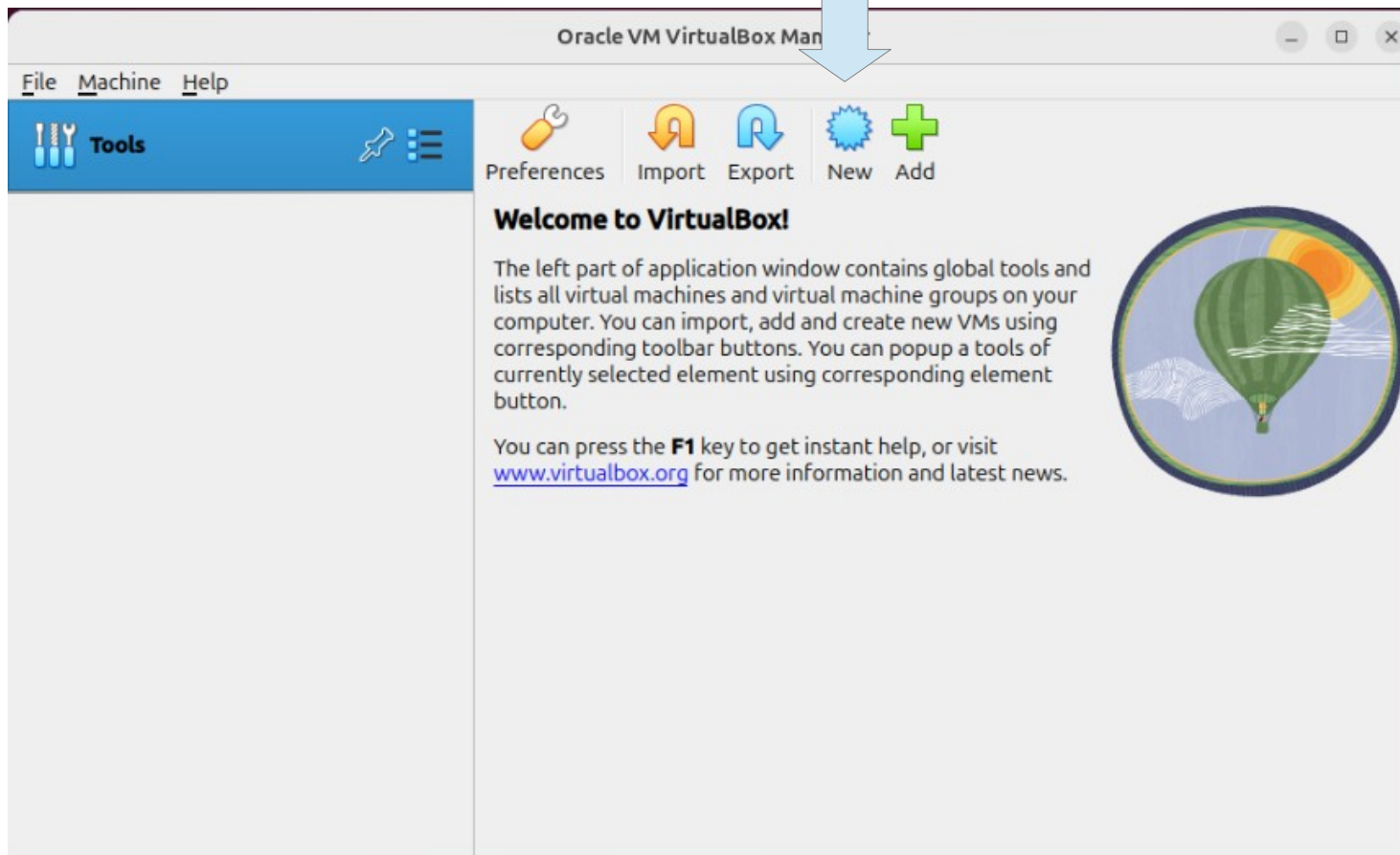
(Windows)

1. Run through Startup Wizard
2. Restart Computer to finish driver installations

WARNING: Virtualization NEEDS to be turned ON in the BIOS

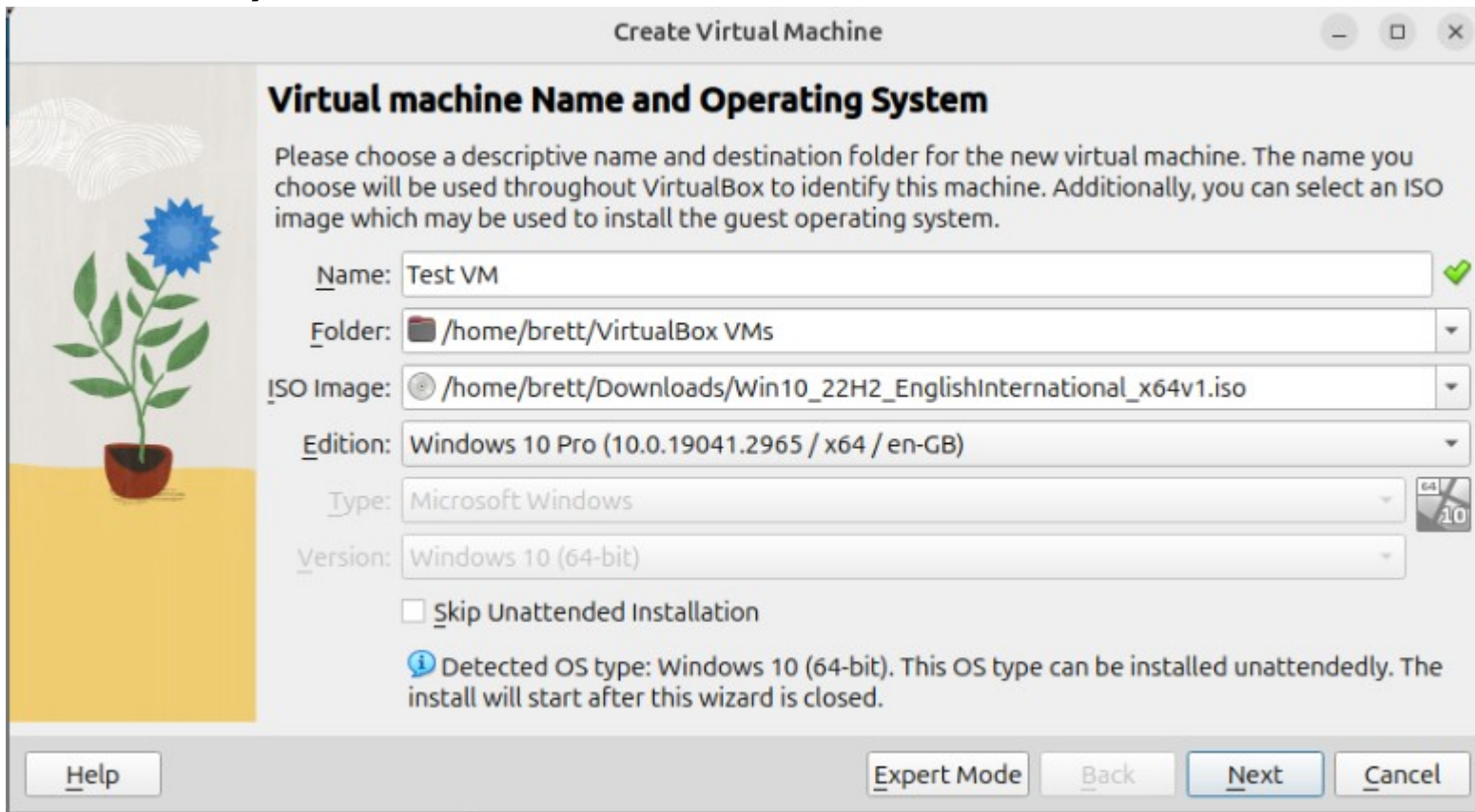
VirtualBox

Add New VM



VirtualBox

Fill out information
(leave “Skip Unattended Installation” unchecked for
easy installation, NOTE: Product key is necessary if on
windows)



Create Virtual Machine

Virtual machine Name and Operating System

Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.

Name: ✓

Folder:

ISO Image:

Edition:

Type:

Version:

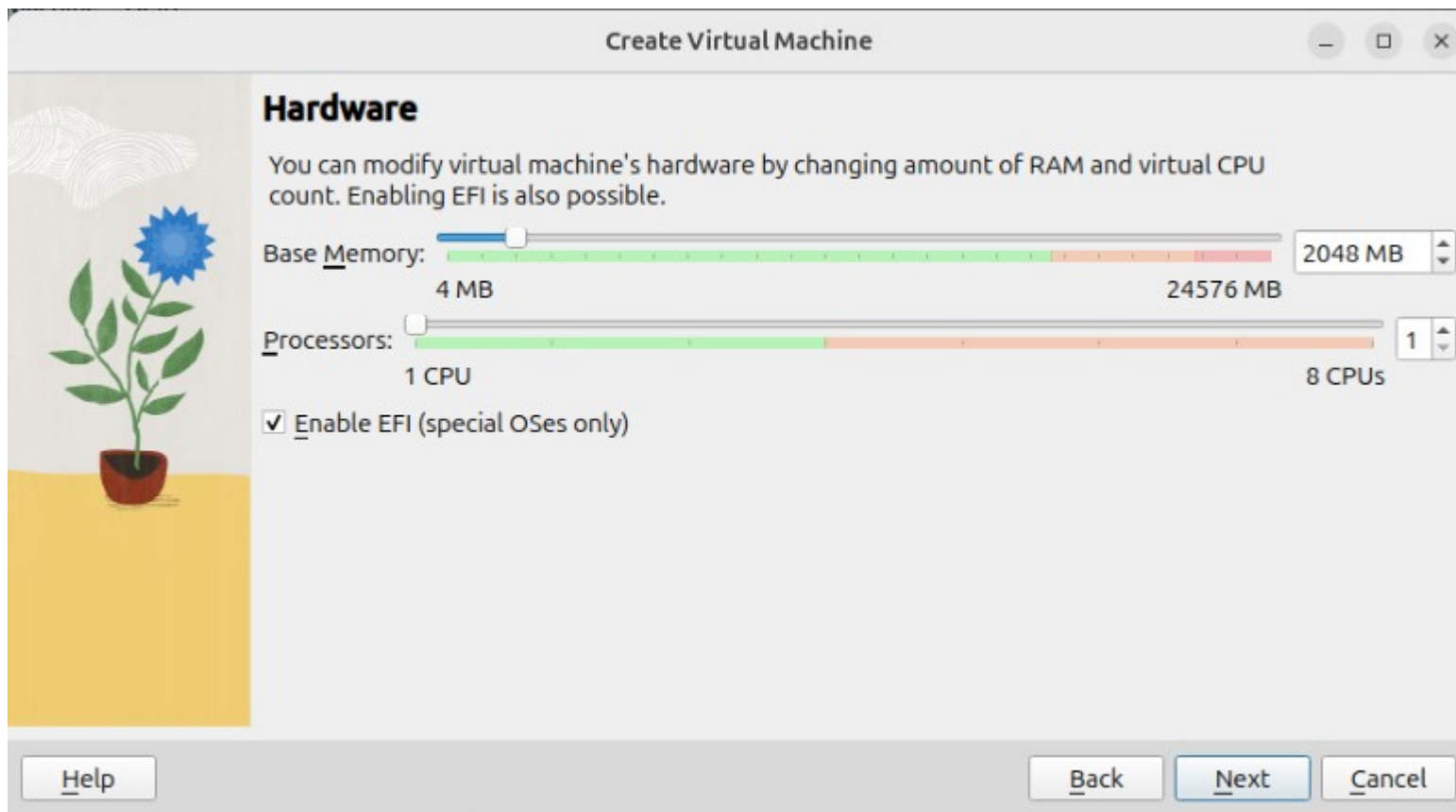
☐ Skip Unattended Installation

Detected OS type: Windows 10 (64-bit). This OS type can be installed unattendedly. The install will start after this wizard is closed.

Buttons: Help, Expert Mode, Back, Next, Cancel

VirtualBox

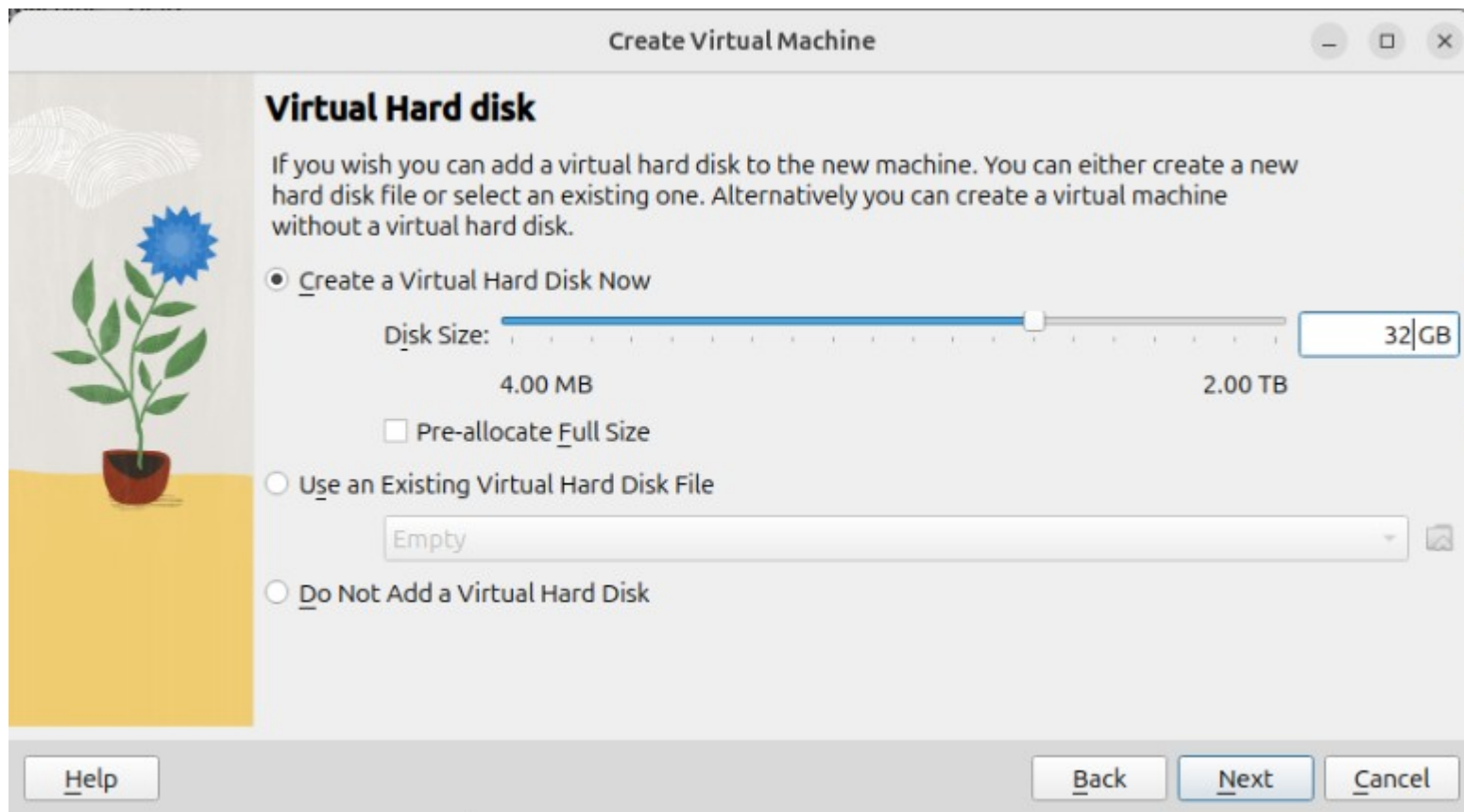
Set Memory Allocation (4 GiB – linux, 8 GiB – Windows)
Allocate vCPUs (2 – linux, 4 Windows)
Enable EFI



VirtualBox

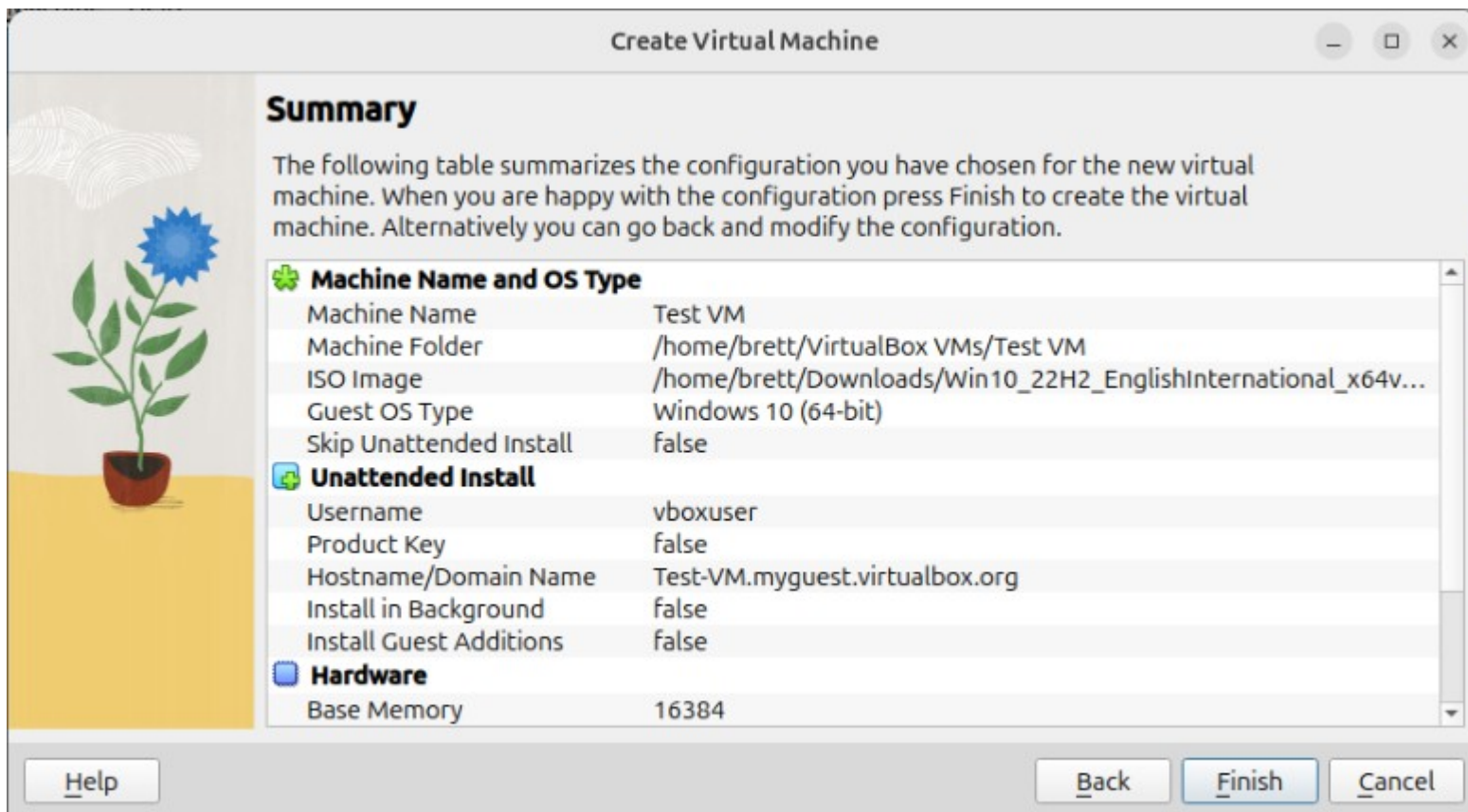
Create Disk (20 GiB – linux, 50 GiB – Windows)

LEAVE “Pre-allocate Full Disk Size” UNCHECKED



VirtualBox

Double check everything and then click “Finish”



Finishing Up

After running through installation

- Check for and run updates
- Install “Guest Drivers” (virtio)
- Check Display Settings

NOTE: GPU Pass-through is NOT possible with VirtualBox

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Questions?

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